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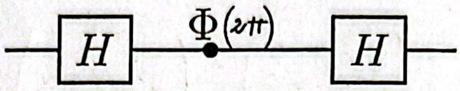
Control 3 **CCQ** **13 Maig 2026**

1. Amb portes phase-shift i Hadamard es pot construir qualsevol porta d'un qubit. Donat el circuit



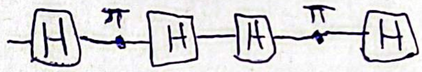
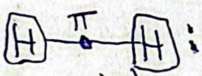
(a) (2 p) Quina matriu (U) en resulta? (expresseu els complexes en forma cartesiana) quina porta és?

$$U = H \cdot \Phi(2\pi) \cdot H = H \cdot \begin{pmatrix} 1 & 0 \\ 0 & e^{i2\pi} \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ e^{i2\pi} & -e^{i2\pi} \end{pmatrix} =$$

$$= \frac{1}{2} \begin{pmatrix} 1+e^{i2\pi} & 1-e^{i2\pi} \\ 1-e^{i2\pi} & 1+e^{i2\pi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$ és la $\mathbb{I}$

✓

(b) (2 p) Trobeu una matriu que implementi $\sqrt{U}$.

Podem pensar en el circuit  tal com vam veure a classe, per tant, $\sqrt{U}$ seria :

$$\sqrt{U} \underset{\uparrow}{=} \frac{1}{2} \begin{pmatrix} 1+e^{i\pi} & 1-e^{i\pi} \\ 1-e^{i\pi} & 1+e^{i\pi} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

aprofitant càlculs anteriors però amb π

Anem a verificar $\sqrt{U} \cdot \sqrt{U} = U$:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = U = \mathbb{I}$$

✓

2. La porta de 2 qubits C-NOT no és imprescindible, podríem emprar-ne d'altres al seu lloc.

(a) (0.5 p) Quina és la matriu (V) phase shift per $\phi = \pi/2$? (1 qubit)

$$V = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

✓

(b) (0.5 p) I la matriu de 2 qubits CONTROL-V?

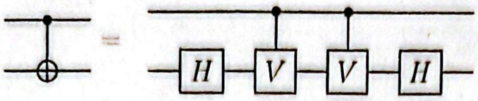
$$\text{Control-}U = \begin{pmatrix} \mathbb{I} & 0 \\ 0 & U \end{pmatrix}$$ llavors substituint U per V:

✓

$$\text{Control-}V = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{array}\right)$$

(c) (3 p) Demostreu la següent equivalència per la porta C-NOT

$$\mathbb{I}\otimes H = \begin{pmatrix}1&0\\0&1\end{pmatrix}\otimes\begin{pmatrix}1&1\\1&-1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1&1&0&0\\1&-1&0&0\\0&0&1&1\\0&0&1&-1\end{pmatrix}$$



$$(\mathbb{I}\otimes H)(C\text{-}V)(C\text{-}V)(\mathbb{I}\otimes H) = (\mathbb{I}\otimes H)(C\text{-}V)\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1&1&0&0\\1&-1&0&0\\0&0&1&1\\0&0&1&-1\end{pmatrix} = (\mathbb{I}\otimes H)\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1&1&0&0\\1&-1&0&0\\0&0&1&1\\0&0&i&-i\end{pmatrix}$$

$$= \frac{1}{\sqrt{2}}\begin{pmatrix}1&1&0&0\\1&-1&0&0\\0&0&1&1\\0&0&1&-1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1&1&0&0\\1&-1&0&0\\0&0&1&1\\0&0&-1&1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2&0&0&0\\0&2&0&0\\0&0&0&2\\0&0&2&0\end{pmatrix} = \begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix} = C\text{-}NOT$$

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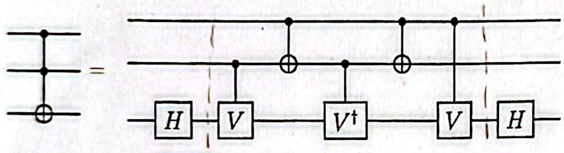
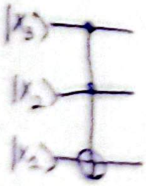
(d) (2 p) Demostreu que podem construir la porta de Toffoli amb el següent circuit.



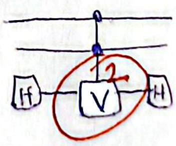
Es pot reduir

Primer entrada



$$\Rightarrow \begin{cases} \overset{x_3 x_2 x_1}{|000\rangle} = |0\,0 \oplus 0\cdot 0\rangle = \overset{x_3 x_2 x_1}{|000\rangle} \\ |001\rangle = |1\,0 \oplus 0\cdot 0\rangle = |001\rangle \\ |010\rangle = |0\,0 \oplus 0\cdot 1\rangle = |010\rangle \\ |011\rangle = |1\,0 \oplus 0\cdot 1\rangle = |011\rangle \\ |100\rangle = |0\,0 \oplus 1\cdot 0\rangle = |100\rangle \\ |101\rangle = |1\,0 \oplus 1\cdot 0\rangle = |101\rangle \\ |110\rangle = |0\,0 \oplus 1\cdot 1\rangle = |111\rangle \\ |111\rangle = |1\,0 \oplus 1\cdot 1\rangle = |110\rangle \end{cases}$$

El circuit també es pot veure com a:



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Per tant sabem que només actua V sobre x_1 quan $x_2 = 1$ i $x_3 = 1$. I la Hadamard deixarà el mateix estat si $x_2 = x_3 \neq 1$.

Per exemple $\overset{x_3 x_2 x_1}{|100\rangle}$: $|10\rangle|0\rangle \xrightarrow{H} |10\rangle\frac{1}{\sqrt{2}}\left[|0\rangle+|1\rangle\right] = \frac{1}{\sqrt{2}}\left[|100\rangle+|101\rangle\right] = \frac{1}{\sqrt{2}}\left[\overset{x_1}{|0\rangle}|10\rangle+\overset{x_1}{|1\rangle}|10\rangle\right]$

torna aplicar H perquè $x_2 \neq x_3$ i no són de les 2

$\longrightarrow \frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}\left[|0\rangle+|1\rangle\right]|10\rangle + \frac{1}{\sqrt{2}}\left[|0\rangle-|1\rangle\right]|10\rangle\right] = \left[\frac{1}{2}|100\rangle+\frac{1}{2}|101\rangle+\frac{1}{2}|100\rangle-\frac{1}{2}|101\rangle\right]$

$= \frac{1}{2}|100\rangle + \frac{1}{2}|100\rangle = |100\rangle$ dona igual que a l'entrada, així igual quan ni x_2 ni x_3 valen 1.

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Titulació

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DNI

2. d)

Quan X_2 i X_3 valen 1:

$$|1\rangle|1\rangle|0\rangle \underset{H}{=} |11\rangle\frac{1}{\sqrt{2}}\left[|0\rangle+|1\rangle\right] = \frac{1}{\sqrt{2}}\left[|110\rangle+|111\rangle\right] = \frac{1}{\sqrt{2}}\left[|0\rangle|11\rangle+|1\rangle|11\rangle\right] =$$

$$\underset{V_{0\,X_1}}{=} \frac{1}{\sqrt{2}}\left[|0\rangle|11\rangle+i|1\rangle|11\rangle\right] \underset{H}{=} \frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}\left[|0\rangle+|1\rangle\right]|11\rangle+\frac{1}{\sqrt{2}}i\left[|0\rangle-|1\rangle\right]|11\rangle\right] =$$

$$= \frac{1}{2}|110\rangle+\frac{1}{2}|111\rangle+\frac{1}{2}i|110\rangle-\frac{1}{2}i|111\rangle$$